PT100 INDUSTRIAL DIGITAL THERMOMETER

Project Progress Log

Semester Project

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This log covers building an industrial digital thermometer using a PT100 temperature sensor (a 330 Ω resistor and the PT100 form a voltage divider that the Arduino Uno reads, and the temperature is shown on a 16×2 LCD). The circuit is soldered on perfboard and we also designed a 3D-printed enclosure for it.

Summary

Date	Milestone
12 Oct	Researched PT100 theory and the circuit, and finalised the circuit layout.
13 Oct	Soldered the circuit on perfboard, but the LCD didn't seem to work.
14 Oct	3D model of the enclosure done. Realised the LCD pin orientation was wrong, so we had to fix it and remake the model.
15 Oct	Desoldered the LCD from the perfboard and used female header pins for it instead. The LCD now shows text.

Day 1 — 12 October

Spent the day reading up on how the PT100 works. It is a platinum resistance sensor, so its resistance goes up almost linearly with temperature (100 Ω at 0 °C), which means we can turn temperature into a voltage using a simple divider. After that we finalised the circuit layout: a 330 Ω resistor in series with the PT100 with the middle point going to pin A0 of the Arduino Uno, a JHD 162A 16×2 LCD, and a potentiometer to adjust the LCD contrast. Used AI assistants to help with the research and the circuit [1], [2].

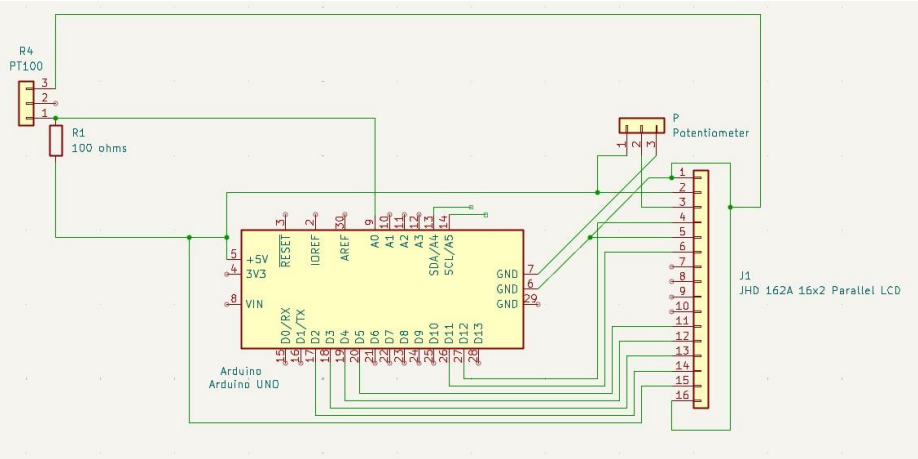


Fig. 1 — Finalised circuit schematic: PT100 and series resistor divider on A0, JHD 162A LCD and contrast potentiometer.

Day 2 — 13 October

Soldered the whole circuit onto the perfboard following the schematic. When we tested it, we ran into a problem: the LCD didn't seem to work, so we couldn't get anything on the display yet.

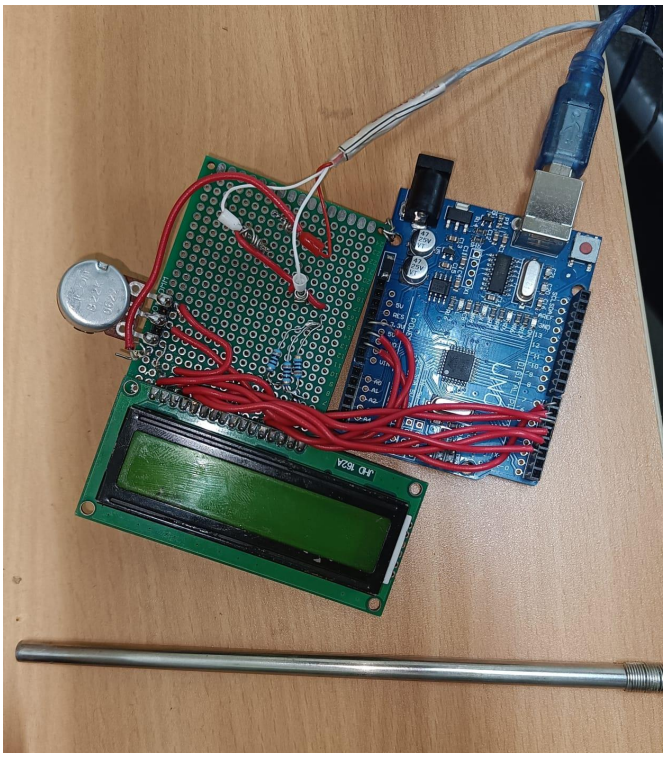


Fig. 2 — Soldered perfboard connected to the Arduino Uno and the LCD, with the PT100 probe alongside.

Day 3 — 14 October

Finished the 3D model of the enclosure: a front panel with a rectangular window for the LCD and some mounting holes, and a base tray with a round hole in one of the side walls and a cutout at one corner. After checking everything carefully against the actual LCD, we noticed we had got the orientation of the LCD pins wrong in the design, so we had to fix that and remake the model.

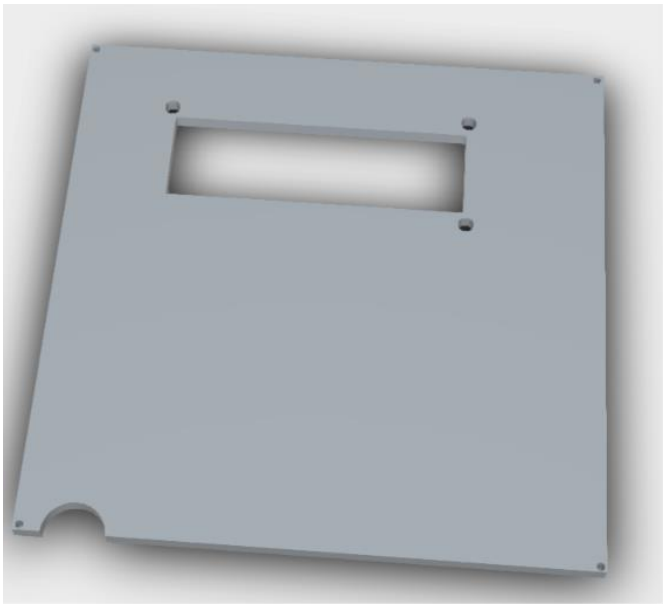


Fig. 3 — Front panel of the enclosure with the LCD window and mounting bosses.

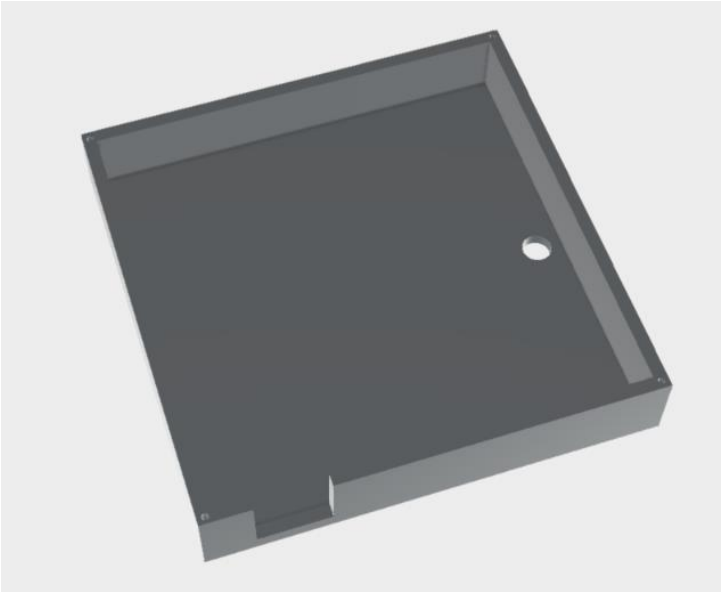


Fig. 4 — Base tray of the enclosure.

Day 4 — 15 October

Today we desoldered the LCD from the perfboard. Instead of soldering it directly again, we used female header pins. The LCD pins just slip into them. With this setup the LCD works and we can finally see text on the display.

References

[1] Anthropic, Claude (Sonnet), Large language model, 2026. [Online]. Available: https://claude.ai

[2] Google, Gemini, Large language model, 2026. [Online]. Available: https://gemini.google.com